Below is a crisp, code-ready blueprint you can hand to engineering. It's organized from product → data → matching → workflow → delivery. Where helpful, I've included concrete schemas, scoring math, and endpoint specs so there's no ambiguity when you start building.

1) Product Objective & Non-Goals

Objective: Help SMB GovCon firms turn past performance + capabilities + NAICS into a prioritized pipeline of forecasted opportunities with auto-generated capture plans, teaming options, and compliance checklists.

Non-Goals (v1):

- No full proposal drafting.
- No paid data feeds or scraping behind logins.
- No CPARS harvesting; allow manual input/import.

2) Primary Users & Roles

- **Founder/BD Lead (Admin):** config, scoring weights, partner lists, export.
- **Capture Manager (Editor):** qualifies opps, launches capture plans, assigns tasks.
- **Analyst (Contributor):** enriches data, drafts artifacts, updates statuses.
- **Executive (Viewer):** dashboards, pipeline health, exports.

3) Core Outcomes (Success Criteria)

- 85%+ precision@20 for recommended opps (human "Qualified" label).
- 30% reduction in time-to-qualification (lead → bid/no-bid).
- Coverage: ≥80% of relevant public agency forecasts loaded and normalized for selected domains.

4) Data Architecture

4.1 Canonical Entities (ERD summary)

- **Contractor**(id, legal_name, dba, UEI, CAGE, small_business_flags{8a,WOSB,SDVOSB,HUBZone}, size_standard_profile, vehicles[], clearances[], facility_locations[])
- **Capability**(id, contractor_id, summary_text, keywords[], NAICS[], PSCs[])
- **PastPerformance**(id, contractor_id, piid, agency_id, title, description, NAICS, PSC, vehicle, value, pop_start, pop_end, role{prime,sub}, cp_qual{optional}, artifacts[])
- **Opportunity**(id, source, source_url, agency_id, bureau_id, office, title, description, keywords[], NAICS[], PSCs[], set_aside, vehicle, contract_type, est_value_min, est_value_max, place_of_performance, pop_est_start, rfi_date, draft_rfp_date, final_rfp_date, status{forecast,draft,final,updated}, contacts{CO,SB}, attachments[], updated_at)
- **Agency**(id, name, parent_id, acronyms[])
- **Vehicle**(id, name, type{IDIQ,BPA,GWAC}, holders[], on_ramp_open{bool})
- **Match**(id, contractor_id, opportunity_id, score_total, score_breakdown{…}, tier{A,B,C}, reasons[], created_at)
- **CapturePlan**(id, opportunity_id, contractor_id, phase{Shape,Pre-RFP,Draft,Final}, checklist[], tasks[], risks[], partners[], artifacts[], status)
- **Partner**(id, name, UEI, vehicles[], certifications[], fit_tags[], contact)

DB: **PostgreSQL** with **pgvector** for embeddings; JSONB for flexible fields.

4.2 Schemas (executable starting point)

Opportunity (SQL)

```
None
CREATE TABLE opportunity (
  id UUID PRIMARY KEY,
  source TEXT NOT NULL,
  source_url TEXT,
  agency_id UUID REFERENCES agency(id),
  bureau_id UUID,
  office TEXT,
  title TEXT NOT NULL,
  description TEXT,
  keywords TEXT[],
  naics TEXT[],    -- 6-digit codes
  pscs TEXT[],     -- 4-char PSC
```

```
  set_aside TEXT,  -- e.g., "8(a)", "WOSB", "SDVOSB", "Small Business"
  vehicle TEXT,    -- canonical name if known
  contract_type TEXT[], -- e.g., ["FFP","T&M"]
  est_value_min NUMERIC,
  est_value_max NUMERIC,
  place_of_performance JSONB, -- {city,state,country,onsite?,remote?}
  pop_est_start DATE,
  rfi_date DATE,
  draft_rfp_date DATE,
  final_rfp_date DATE,
  status TEXT CHECK (status IN ('forecast','draft','final','updated')),
  contacts JSONB, -- {co:{name,email,phone}, sb:{name,email,phone}}
  attachments JSONB,
  updated_at TIMESTAMP DEFAULT now()
);
```

Contractor Profile (SQL)

```
None
CREATE TABLE contractor (
  id UUID PRIMARY KEY,
  legal_name TEXT,
  dba TEXT,
  uei TEXT UNIQUE,
  cage TEXT,
  small_business_flags JSONB, --
{"8a":true,"WOSB":false,"SDVOSB":false,"HUBZone":true}
  size_standard_profile JSONB, -- {"541511":"$34.0M","541512":"$34.0M",...}
  vehicles TEXT[],             -- normalized vehicle names
  clearances TEXT[],           -- e.g., "Secret","Top Secret"
  facility_locations JSONB     -- [ {city,state,zip}, ... ]
);
```

Vector Columns

```
None
-- pgvector
ALTER TABLE opportunity ADD COLUMN desc_embedding VECTOR(1536);
ALTER TABLE capability ADD COLUMN text_embedding VECTOR(1536);
```

5) Ingestion & Normalization

5.1 Ingestion Adapters

- **Forecast adapters** (per agency): HTML/PDF/CSV → normalized Opportunity rows.
- **Past performance**: CSV import (PIID, agency, NAICS, role, value, dates, text); optional manual entry.
- **Contractor profile**: quick form + CSV import for NAICS, vehicles, flags, clearances.

5.2 Normalization Rules

- **NAICS**: enforce 6-digit; include roll-up to 3/4-digit families for similarity features.
- **PSC**: uppercase, 4-char.
- **Agency**: canonicalize to parent/bureau using normalized table; keep aliases.
- **Set-Aside**: map synonyms to canonical types; allow multiple where forecasts are ambiguous.
- **Vehicle**: canonical list; map common variants (e.g., “Alliant 2” → “Alliant 2”).

5.3 Entity Resolution

- Contractors by **UEI/CAGE**; Agencies by canonical id; Vehicles by normalized name.

5.4 Data Quality

- Required fields: title, agency_id, at least one of {NAICS, PSC, description}.
- Staleness tag if updated_at > 180 days.
- Provenance: source, source_url, parser_version.

6) Matching & Scoring

6.1 Features (per contractor × opportunity)

Gating (0/1, hard filters that can down-weight or block):

- Vehicle access required? If yes and contractor lacks seat → score cap at 0.60 (recommend teaming).
- Set-aside eligibility conflict? (e.g., SDVOSB required) → cap at 0.50 (teaming).
- Clearance required and not held → cap at 0.50.
- SBA size standard fail for primary NAICS → cap at 0.70.

Signals (0-1 each, weighted):

- **NAICS Match**: exact(1.0), family(0.7), adjacent(0.4).
- **PSC Match**: exact(1.0), related(0.6).
- **Text Similarity** (embeddings): cosine(capability.summary, opportunity.description/keywords).
- **Agency Affinity**: any past perf at that agency family (parent→bureau): parent(1.0), sibling bureau(0.7), similar mission cluster(0.5).
- **Role Experience**: prime experience in same contract_type/vehicle family (0.2-0.6).
- **Value Fit**: within historical award band ±X% (0.2-0.6).
- **Geo Feasibility**: presence near PoP or remote allowed (0.2-0.5).
- **Timing**: lead time until draft/final RFP vs internal cycle time (0.2-0.6).
- **Certifications**: match to set-aside, CMMC/NIST when indicated (0.2-0.4).

Default Weights (editable per account):

```None
w = {
  naics: 0.20,
  psc: 0.10,
  text: 0.25,
  agency: 0.15,
  role_vehicle: 0.05,
  value_fit: 0.05,
  geo: 0.05,
  timing: 0.10,
  certs: 0.05
}
```

6.2 Scoring Formula

```
None
raw = Σ (w_i * feature_i)                 // all features scaled 0–1
cap = min(gates...)                       // gates ∈ (0,1], e.g., 0.60
score_total = raw * cap
tier = A if score_total ≥ 0.75
       B if 0.55–0.74
       C if 0.40–0.54
       else "Ignore"
reasons = top 3 positive + top 3 blockers (human-readable)
```

6.3 Pseudocode (deterministic & debuggable)

```
None
def compute_match_score(contractor, opp, weights):
    gates = []
    if opp.vehicle and opp.vehicle in REQUIRE_VEHICLE_ACCESS:
        gates.append(0.60 if opp.vehicle not in contractor.vehicles else
1.0)
    if opp.set_aside in {'8(a)','WOSB','SDVOSB','HUBZone'}:
        gates.append(0.50 if not
contractor.small_business_flags.get(opp.set_aside, False) else 1.0)
    if requires_clearance(opp) and not has_clearance(contractor, opp):
        gates.append(0.50)
    if primary_naics := primary_naics_of(opp):
        if not passes_size_standard(contractor, primary_naics):
            gates.append(0.70)
    cap = min(gates) if gates else 1.0

    f = {}
    f['naics'] = naics_similarity(contractor.capabilities.naics, opp.naics)
# 0-1
    f['psc']   = psc_similarity(contractor.capabilities.pscs, opp.pscs)
# 0-1
    f['text']  = cosine(emb(contractor.capabilities.text), emb(opp.text))
# 0-1
    f['agency']= agency_affinity(contractor.past_performance, opp.agency_id)
# 0-1
```

```
    f['role_vehicle'] = role_vehicle_experience(contractor, opp)
# 0-1
    f['value_fit'] = value_band_fit(contractor.past_performance,
opp.est_value_min, opp.est_value_max)
    f['geo']    = geo_feasibility(contractor.facility_locations,
opp.place_of_performance)
    f['timing'] = timing_readiness(opp, contractor.internal_cycle_days)
    f['certs']  = certs_alignment(contractor, opp)

    raw = sum(weights[k]*f[k] for k in weights)
    score = raw * cap
    tier = 'A' if score>=0.75 else 'B' if score>=0.55 else 'C' if
score>=0.40 else 'Ignore'
    reasons = explain_top_signals(f, cap, contractor, opp)
    return score, tier, reasons, f, cap
```

7) Workflow & Deliverables

7.1 Qualification Pipeline (states)

New → Review → Qualified → Capture → Bid/No-Bid → Submitted/Abandoned → Award/Loss

State transitions require CEPA checks (below) and short forms.

7.2 CEPA Pre-Bid Checklist (embedded)

- **Compliance fit:** set-aside eligibility, primary NAICS size standard, vehicle access.
- **Strategic fit:** agency affinity score ≥ threshold, capability/text similarity ≥ threshold.
- **Timing fit:** ≥ N days until Draft/Final RFP.
- **Resourcing:** named Capture Lead, Writer, Pricing, Tech SME.
- **Intelligence:** at least one contact (CO/OSDBU/PMO) and two insight bullets.
- **Risk:** top 3 risks with mitigation owners.

If any fail → remain in Review or auto-suggest teaming.

7.3 Auto-Generated Capture Plan (upon "Qualified")

- **Phases & Tasks** (date-backed from draft/final RFP):
  - *Shape:* stakeholder mapping, Q&A topics, capability gaps, partner outreach.
  - *Pre-RFP:* win themes, solution sketch, discriminators, past-perf mapping.
  - *Draft RFP:* section L/M compliance matrix stub, color-team schedule.
  - *Final:* final compliance matrix, pricing inputs, forms checklist.
- **Artifacts:** capture brief (1-pager), partner matrix, compliance matrix, Q&A list.
- **Owners & due dates** auto-assigned from role defaults.

7.4 Teaming Recommendations

- If vehicle/set-aside gate fails: list **Partner** entities with needed seat/flag and high text/NAICS overlap; include contact info and "why" tags.

8) UX Surfaces

- **Home / Dashboard:** A/B/C tiers, aging, upcoming dates, bottlenecks.
- **Discover:** faceted search on normalized forecast repo.
- **Matches:** contractor-specific ranked list with score breakdown + reasons.
- **Opportunity Detail:** canonical data, blockers, suggested partners, CEPA status, create Capture Plan.
- **Capture Plan:** phases, tasks, checklists, artifacts, notes, exports.
- **Partner Matrix:** filters by vehicle, set-aside, agency performance, similarity.
- **Reports/Exports:** CSV/Excel of pipeline, match breakdowns, CEPA audit trail.

9) APIs (v1, REST)

```None
POST   /v1/contractors            # create/update profile
POST   /v1/contractors/{id}/capabilities
POST   /v1/contractors/{id}/past_performance/import  # CSV
POST   /v1/opportunities/import    # bulk load forecasts (per-adapter)
GET    /v1/opportunities/search?q=&filters=...
POST   /v1/match/run               # {contractor_id, filters?} -> matches[]
GET    /v1/match/{contractor_id}?opp_id=...
POST   /v1/capture_plans           # create from opp+contractor
GET    /v1/capture_plans/{id}
POST   /v1/partners/import
GET    /v1/partners/search
```

Match response (example):


```None
{
  "opportunity_id": "…",
  "score_total": 0.82,
  "tier": "A",
  "cap": 1.0,
  "breakdown":
{"naics":0.9,"psc":0.8,"text":0.78,"agency":1.0,"timing":0.6,...},
  "reasons": ["Exact NAICS 541511","Strong capability similarity","Parent
agency past performance"],
  "blockers": []
}
```


10) Engineering Plan (Phases)

Phase 0 – Foundations

- Postgres + pgvector; service scaffolding (FastAPI); auth (JWT).
- Data model migration; seed Agency/Vehicle canonical tables.

Phase 1 – Forecast Repo

- Build 3–5 ingestion adapters (CSV, HTML, PDF-to-text).

- Normalization, dedupe, provenance; admin upload UI.

Phase 2 – Matching Engine

- Embedding pipeline; gating + weighted scoring; explainability payload.
- Configurable weights per contractor; evaluation harness with labeled set.

Phase 3 – Capture Plans

- CEPA checks; task templates; assignments; artifact templates.
- Partner recommendation MVP.

Phase 4 – UX & Exports

- Dashboards, filters; Excel/CSV exports; audit trail.

Phase 5 – Integrations (optional)

- Google/Outlook Calendar sync for milestones.
- CRM push (e.g., HubSpot deal creation) via webhook.

11) Data Science & Evaluation

- **Ground Truth:** small labeled set (100–300 opps) marked Qualified/Not by capture managers.
- **Metrics:** precision@K, recall@K, ROC-AUC, time-to-qualification.
- **Offline Tuning:** grid search on weights; keep per-account profiles (e.g., “Services-heavy,” “R&D-leaning”).

12) Security & Governance

- RBAC: Admin/Editor/Viewer; row-level scoping by tenant_id.
- Audit log on match runs, weight changes, and capture decisions.
- PII minimization; encryption at rest (Postgres TDE or disk-level) and in transit (TLS).

- Data retention policy (e.g., 24-month rolling forecast storage).

13) Ops & Reliability

- Ingestion workers (Prefect/Celery) with retries; dead-letter queue.
- Freshness SLA badges on opportunities (90/180-day thresholds).
- Monitoring: ETL success rate, parser errors, match latency, embedding queue depth.

14) Files, Templates, and Exports (ready for content)

- **Capture Brief (1-pager)**: opp summary, score, reasons, blockers, win themes, next 3 actions.
- **Compliance Matrix Stub**: auto-generated from opp text (if RFP present) else placeholder.
- **Partner Matrix CSV**: partner, vehicle, set-aside, affinity score, contact.

15) Example: CSV Specs

Past Performance Import (past_performance.csv)

```
None
piid,agency,bureau,title,description,naics,psc,vehicle,role,value,pop_start,pop_end,prime_partner
```

Forecast Import (opportunities.csv)

```
None
source,source_url,agency,bureau,office,title,description,keywords,naics,pscs,set_aside,vehicle,contract_type,est_value_min,est_value_max,place_city,place_state,pop_est_start,rfi_date,draft_rfp_date,final_rfp_date,co_name,co_email,co_phone,sb_name,sb_email,sb_phone
```

16) Seed Algorithms (helpers)

- **NAICS Similarity:** exact→1.0; same 4-digit family→0.7; same 3-digit family→0.5; else 0.
- **PSC Relatedness:** curated mapping table; fallback to embedding similarity of PSC descriptions.
- **Agency Affinity:** parent bureau graph (adjacency list) with decay.
- **Timing Readiness:** logistic curve on days_until(draft/final) vs internal cycle time.
- **Value Fit:** compare opp est_value_mid to median(contractor past_perf values).

17) Risks & Mitigations

- **Forecast volatility:** Store updated_at, surface staleness; diff viewer for changes.
- **Adapter drift:** Version adapters; add canary validation and sample snapshots.
- **Cold-start contractors:** default to NAICS+text only; offer quick capability intake.

18) What Engineering Builds First (MVP cut)

1. Forecast ingestion (CSV + one HTML adapter) → normalized repo.
2. Contractor intake (form + CSV) for NAICS, capabilities text, vehicles, flags.
3. Matching service with gating, weights, explainability; Matches UI with tiers.
4. “Create Capture Plan” button with CEPA checklist and task template; CSV/Excel export.

If you want, I can generate:

- The initial **Postgres migration files**,
- A **FastAPI skeleton** with the /match/run endpoint implemented exactly as specified, and
- The **CEPA checklist JSON** + default capture task templates.

Say the word and I’ll produce those artifacts in a single pass.